

Module 8: Cellular Genetics — Study Questions

1. Name the three stages of interphase (G_1 , S, G_2). What major event occurs during each?
2. What happens during S phase? Why is it essential before cell division?
3. After S phase, are the two copies of a chromosome called homologs or sister chromatids? How can you tell the difference?
4. List the four phases of mitosis in order (PMAT). Briefly describe what happens during each.
5. What structures separate during anaphase of mitosis?
6. Compare cytokinesis in animal cells (cleavage furrow) and plant cells (cell plate). Why do they differ?
7. At the end of mitosis and cytokinesis, how many daughter cells are produced? Are they diploid or haploid? Are they genetically identical to the parent cell?
8. What are the three main cell cycle checkpoints (G_1 , G_2 , M)? What is evaluated at each?
9. Where does meiosis occur in humans? What is the purpose of meiosis?
10. How many rounds of division occur in meiosis? How many daughter cells are produced?
11. What structures separate during Anaphase I? How does this differ from what separates in Anaphase II?
12. Describe synapsis and crossing over. During which phase do they occur? Why are they important?
13. What is independent assortment? How does random alignment of homologs at Metaphase I contribute to genetic diversity?

14. Calculate: In humans ($2n = 46$, or 23 pairs), how many unique gamete combinations are possible from independent assortment alone?
15. Complete a comparison table with these features: number of divisions, number of daughter cells, ploidy of daughter cells, genetic identity to parent, purpose.
16. Does crossing over occur in mitosis? Why or why not?
17. Why is it critical that gametes are haploid? What would happen if gametes were diploid?
18. Name the three sources of genetic variation that occur during sexual reproduction. Which two occur during meiosis?
19. How does random fertilization further increase genetic diversity beyond what crossing over and independent assortment produce?
20. Explain why siblings (from the same parents) are not genetically identical to each other.
21. What is nondisjunction? During which stage(s) of meiosis can it occur?
22. If nondisjunction occurs during Meiosis I, how many gametes are abnormal? What if it occurs during Meiosis II?
23. What is aneuploidy? Give one example of an aneuploid condition in humans and its cause.
24. What are cell cycle checkpoints? Why are they important for preventing cancer?
25. What is the role of the p53 tumor suppressor protein? Why are mutations in p53 so common in cancers?
26. Distinguish between a proto-oncogene and an oncogene. What type of mutation converts one to the other?
27. A mutation inactivates the Rb (retinoblastoma) protein. Predict the effect on cell division.

28. Normal cells divide approximately 50 times before senescence (Hayflick limit). Cancer cells are effectively "immortal." How does telomerase reactivation contribute to this?